

Singular Perturbations, Moments and Descriptor Systems: Linear

Yoonjae Hwang

September 23, 2025

1 Linear Systems - Thinking about explicit solutions and moments

Consider the following linear time-invariant, SISO system:

$$\begin{aligned}\dot{x} &= Ax + Bu, \\ y &= Cx,\end{aligned}\tag{1}$$

$$\begin{aligned}\dot{w} &= Sw, \\ u &= Lw,\end{aligned}\tag{2}$$

Given initial conditions $x(0) = x_0$ and $w(0) = w_0$, the solution to the system is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}BL e^{S\tau}w_0 d\tau.\tag{3}$$

The output of the system is given by

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}BL e^{S\tau}w_0 d\tau.\tag{4}$$

If $\sigma(A) \cup \sigma(S) = \emptyset$, then there exists a unique solution Π to the Sylvester equation

$$A\Pi + BL = \Pi S.\tag{5}$$

Using this, we can substitute an expression for BL into the integral term for $x(t)$:

$$\begin{aligned}x(t) &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}(\Pi S - A\Pi)e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}\Pi S e^{S\tau}w_0 d\tau - \int_0^t e^{A(t-\tau)}A\Pi e^{S\tau}w_0 d\tau \\ &= e^{At}x_0 + (\Pi e^{St} - e^{At}\Pi)w_0.\end{aligned}\tag{6}$$

Forward Moments are defined to be

$$M_g = C\Pi\tag{7}$$

2 Problem Setup

Consider the following linear singularly perturbed system:

$$\begin{aligned}\dot{x}_1(t) &= A_{11}x_1(t) + A_{12}x_2(t) + B_1u(t), \\ \varepsilon\dot{x}_2(t) &= A_{21}x_1(t) + A_{22}x_2(t) + B_2u(t),\end{aligned}\tag{8}$$

where $x_1(t) \in \mathbb{R}^{n_1}$, $x_2(t) \in \mathbb{R}^{n_2}$, $u(t) \in \mathbb{R}^m$, and $0 < \varepsilon \ll 1$.

You can write this as a descriptor system:

$$E_1\dot{x} = Ax + Bu\tag{9}$$

where

$$E_1 = \begin{bmatrix} I_{n_1} & 0 \\ 0 & \varepsilon I_{n_2} \end{bmatrix}, \quad A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \quad (10)$$

The input signal is also a linear singularly perturbed system:

$$\begin{aligned} \dot{w}_1 &= S_{11}w_1 + S_{12}w_2 \\ \varepsilon \dot{w}_2 &= S_{21}w_1 + S_{22}w_2 \end{aligned} \quad (11)$$

where $w_1(t) \in \mathbb{R}^{r_1}$, $w_2(t) \in \mathbb{R}^{r_2}$, and $0 < \varepsilon \ll 1$. The input to the system is given by $u(t) = L_1w_1(t) + L_2w_2(t)$.

The input signal can also be written as a descriptor system like so:

$$E_2\dot{w} = Sw \quad (12)$$

where

$$E_2 = \begin{bmatrix} I_{r_1} & 0 \\ 0 & \varepsilon I_{r_2} \end{bmatrix}, \quad S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}. \quad (13)$$

The output of the system is given by

$$y(t) = C_1x_1(t) + C_2x_2(t). \quad (14)$$

3 Weierstrass Canonical Form

For a descriptor system of the form $E\dot{x} = Ax + Bu$, the matrix pencil (A, E) is said to be regular if there exists a complex number s such that $\det(sE - A) \neq 0$. If the matrix pencil (A, E) is regular, then there exist nonsingular matrices P and Q such that

$$PEQ = \begin{bmatrix} I_{n_1} & 0 \\ 0 & N \end{bmatrix}, \quad PAQ = \begin{bmatrix} J & 0 \\ 0 & I_{n_2} \end{bmatrix}, \quad (15)$$

where J is in Jordan canonical form, and N is a nilpotent matrix.

Let's assume (A, E_1) and (S, E_2) are regular. So for the singular perturbation system, we have

$$PE_1Q = \begin{bmatrix} I_{n_1} & 0 \\ 0 & N \end{bmatrix}, \quad PAQ = \begin{bmatrix} J_A & 0 \\ 0 & I_{n_2} \end{bmatrix}, \quad (16)$$

and

$$WE_2T = \begin{bmatrix} I_{r_1} & 0 \\ 0 & M \end{bmatrix}, \quad WST = \begin{bmatrix} J_S & 0 \\ 0 & I_{r_2} \end{bmatrix}, \quad (17)$$

where J_A and J_S are in Jordan canonical form, and N and M are nilpotent matrices.

Question: Can we find P and Q for this? How does this relate to A_{22} being Hurwitz?

We can rewrite the system with a coordinate transformation $x = Q\tilde{x}$ and $w = T\tilde{w}$, with $\tilde{x} = \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}$ and $\tilde{w} = \begin{bmatrix} \tilde{w}_1 \\ \tilde{w}_2 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} I_{n_1} & 0 \\ 0 & N \end{bmatrix} \begin{bmatrix} \dot{\tilde{x}}_1(t) \\ \dot{\tilde{x}}_2(t) \end{bmatrix} &= \begin{bmatrix} J_A & 0 \\ 0 & I_{n_2} \end{bmatrix} \begin{bmatrix} \tilde{x}_1(t) \\ \tilde{x}_2(t) \end{bmatrix} + \tilde{B}(L_1\tilde{w}_1(t) + L_2\tilde{w}_2(t)), \\ \begin{bmatrix} I_{r_1} & 0 \\ 0 & M \end{bmatrix} \begin{bmatrix} \dot{\tilde{w}}_1(t) \\ \dot{\tilde{w}}_2(t) \end{bmatrix} &= \begin{bmatrix} J_S & 0 \\ 0 & I_{r_2} \end{bmatrix} \begin{bmatrix} \tilde{w}_1(t) \\ \tilde{w}_2(t) \end{bmatrix}, \\ y(t) &= \tilde{C}_1\tilde{x}_1(t) + \tilde{C}_2\tilde{x}_2(t), \end{aligned} \quad (18)$$

References